

Model	Run param	Metrics
Base model	(on old commit) <pre> srun python run_study.py -- targets Count_EX1 Count_EX1 \ --study_name water_modelling \ --study_count 100 \ --onehot-encoding \ --discard-features "SEN,L_wave_ht[mm]" \ --subsample-shap (above is incorrect!) srun python run_study_ev.py --targets Count_EX1 Count_EX2 \ --study-name base_model_try3 \ --study_count 100 \ --onehot-encoding \ --discard-features "" \ --subsample-shap </pre>	Ex1: <pre> "mean_absolute_error": 2.9448960279820993, "mean_absolute_percent_err or": 8.96396995722928, "r_squared": 0.9369697341354437, "mean_squared_error": 120.49129975085198, "root_mean_squared_error": 10.976852907407112, Ex2: "mean_absolute_error": 2.757123034511648, "mean_absolute_percent_err or": 8.419863595468433, "r_squared": 0.9318268309370233, "mean_squared_error": 104.4407562901402, "root_mean_squared_error": 10.219626034750009, </pre>
Data cleaning only Waterflow -> categorical Fixing misread data from filename (waterflow, airflow, CF confused, SEN)	<pre> 08-17-data-cleaned_%j.out srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name 08-17-data- cleaned \ --study_count 100 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread </pre>	<pre> Ex1: "mean_absolute_error": 3.3410444372515817, "mean_absolute_percent_err or": 10.074013072551368, "r_squared": 0.9251522661518765, "mean_squared_error": 144.97590208033012, "root_mean_squared_error": 12.040593925564059, "root_mean_squared_log_err or": 0.22320586457187086, Ex2: "mean_absolute_error": </pre>

		2.8623989611452547, "mean_absolute_percent_error": 8.700897823703144, "r_squared": 0.9344439972053219, "mean_squared_error": 102.54118610254429, "root_mean_squared_error": 10.126262197995088, "root_mean_squared_log_error": 0.19547166491152781,
Data cleaning ignore time* multithread	srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name data_cleaned_ignore_time_o nly \ --study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --discard-features "time[s]" \ 	EX1: "mean_absolute_error": 4.007190891142333, "mean_absolute_percent_error": 12.186470184536622, "r_squared": 0.895574266614676, "mean_squared_error": 202.26684389212085, "root_mean_squared_error": 14.222054840708527, EX2: "mean_absolute_error": 3.5803086304874374, "mean_absolute_percent_error": 10.977055984831239, "r_squared": 0.9000206220020979, "mean_squared_error": 156.38543487480246, "root_mean_squared_error": 12.505416221573853,
Data cleaning ignore time sen-geometrical*multithread	srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name data_cleaned_ignore_time_g eo \ 	EX1: "mean_absolute_error": 4.091652064773601, "mean_absolute_percent_error": 12.557071814985237,

	<pre>--study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --discard-features "time[s]" \ --sen-geometrical \</pre>	<pre>"r_squared": 0.8959504219272167, "mean_squared_error": 201.53825195013147, "root_mean_squared_error": 14.19641687011661, EX2: "mean_absolute_error": 3.5916849385786116, "mean_absolute_percent_err or": 10.940753303721602, "r_squared": 0.9022315002481955, "mean_squared_error": 152.92723016404216, "root_mean_squared_error": 12.366374980730697,</pre>
Data cleaning ignore time sen geometrical mould-pos	<pre>sr run python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name data_cleaned_ignore_time_g eo_mld \ --study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \</pre>	<pre>EX1: "mean_absolute_error": 2.0823999456381457, "mean_absolute_percent_err or": 5.569616036571325, "r_squared": 0.9670459667808043, "mean_squared_error": 62.879203265691764, "root_mean_squared_error": 7.9296408030686845, EX2: "mean_absolute_error": 1.8853451436247788, "mean_absolute_percent_err or": 5.249922424615964, "r_squared": 0.9724187554633228, "mean_squared_error": 42.39216742481372, "root_mean_squared_error": 6.510926771575128,</pre>

Data cleaning ignore time sen geometrical lag 3	<pre> srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name 0818_time_geo_3 \ --study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --discard-features "time[s]" \ --sen-geometrical \ --lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s],AN_3_RQ[m/s],AN_4_RR[m /s],ML_LL[mm],ML_LQ[mm], ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 3 \ </pre>	EX1: <pre> "mean_absolute_error": 3.7251680445745503, "mean_absolute_percent_err or": 11.279962116438833, "r_squared": 0.9054239467161043, "mean_squared_error": 180.74974537972665, "root_mean_squared_error": 13.444320190315562, EX2: "mean_absolute_error": 3.2285105811656885, "mean_absolute_percent_err or": 9.876045548427886, "r_squared": 0.9161479911115012, "mean_squared_error": 131.13551671867015, "root_mean_squared_error": 11.451441687345316, </pre>
Data cleaning ignore time sen geometrical lag 5	<pre> srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name data_cleaned_ignore_time_g eo_5 \ --study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --discard-features "time[s]" \ --sen-geometrical \ --lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s],AN_3_RQ[m/s],AN_4_RR[m /s],ML_LL[mm],ML_LQ[mm], </pre>	EX1: <pre> "mean_absolute_error": 3.6399448821848694, "mean_absolute_percent_err or": 10.781075045986247, "r_squared": 0.9092291886381023, "mean_squared_error": 175.31815792629436, "root_mean_squared_error": 13.240776333972807, EX2: "mean_absolute_error": 3.276396482044914, "mean_absolute_percent_err </pre>

	ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 5 \	or": 9.939085595739575, "r_squared": 0.9167133582279033, "mean_squared_error": 132.80832159593444, "root_mean_squared_error": 11.524249285568862,
Data cleaning ignore time sen geometrical lag 10	sruntime python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name data_cleaned_ignore_time_g eo_10 \ --study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --discard-features "time[s]" \ --sen-geometrical \ --lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s],AN_3_RQ[m/s],AN_4_RR[m /s],ML_LL[mm],ML_LQ[mm], ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 10 \	EX1: "mean_absolute_error": 3.573221487111727, "mean_absolute_percent_err or": 11.292755651986216, "r_squared": 0.9217188331351277, "mean_squared_error": 151.5428269165969, "root_mean_squared_error": 12.310273226723966, EX2: "mean_absolute_error": 3.301125828789674, "mean_absolute_percent_err or": 10.100260824919904, "r_squared": 0.9185222162829304, "mean_squared_error": 131.05257367384166, "root_mean_squared_error": 11.447819603480903,
Data cleaning ignore time sen geometrical lag 10 mould-pos	sruntime python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name data_cleaned_ignore_time_g eo_mld_10 \ --study_count 5 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \	EX1: "mean_absolute_error": 2.0445126856732045, "mean_absolute_percent_err or": 5.5154070640832735, "r_squared": 0.9682458330530643, "mean_squared_error": 60.300187053436055,

	--clogging-factors \ --multithread \ --discard-features "time[s]" \ --sen-geometrical \ --lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s]],AN_3_RQ[m/s],AN_4_RR[m /s],ML_LL[mm],ML_LQ[mm], ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 10 \ --mould-position \ 	"root_mean_squared_error": 7.765319507492016, EX2: "mean_absolute_error": 1.8282644928490148, "mean_absolute_percent_err or": 5.080602707098262, "r_squared": 0.973585395040737, "mean_squared_error": 39.5187683693301, "root_mean_squared_error": 6.286395498958851,
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Model	Argument	Metrics
Data cleaning ignore time sen geometrical mould-pos clean data	srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name final_noclog \ --study_count 100 \ --onehot-encoding \ --data-directory noClog \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \ 	EX1: "mean_absolute_error": 2.2807976818426354, "mean_absolute_percent_err or": 5.7413912607600714, "r_squared": 0.962908640294317, "mean_squared_error": 72.6376767571107, "root_mean_squared_error": 8.522774005986003, EX2: "mean_absolute_error": 2.1445507422546166, "mean_absolute_percent_err or": 5.421463271144499, "r_squared": 0.9677711953035881, "mean_squared_error": 57.35305090457031, "root_mean_squared_error": 7.573179708984219,

Data cleaning ignore time sen geometrical mould-pos sen7 & clogged	<pre> srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name final_clog \ --study_count 100 \ --onehot-encoding \ --data-directory cloggedData \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \ </pre>	EX1: <pre> "mean_absolute_error": 2.018628779901002, "mean_absolute_percent_err or": 5.133096611897589, "r_squared": 0.9656532963671585, "mean_squared_error": 50.954111181535765, "root_mean_squared_error": 7.1382148455713885, EX2: "mean_absolute_error": 1.7532117926606896, "mean_absolute_percent_err or": 4.819747839712123, "r_squared": 0.9675981132455568, "mean_squared_error": 37.25036364392273, "root_mean_squared_error": 6.103307598665066, </pre>
Data cleaning ignore time sen geo mld-pos clogged SEN with time lag 10	<pre> srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name clogged_lag \ --study_count 100 \ --onehot-encoding \ --data-directory cloggedData \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \ --feature-lag-amount 10 \ </pre>	EX1: <pre> "mean_absolute_error": 2.0205477978668926, "mean_absolute_percent_err or": 5.100554425071913, "r_squared": 0.965313661168552, "mean_squared_error": 51.45796767547922, "root_mean_squared_error": 7.173420918604959, EX2: "mean_absolute_error": 1.7606299066076052, </pre>

		<pre>"mean_absolute_percent_error": 4.851071321505247, "r_squared": 0.9677303523245184, "mean_squared_error": 37.09833688645057, "root_mean_squared_error": 6.0908404088804176,</pre>
Data cleaning ignore time sen geo mld-pos clogged SEN with time lag 10 one study	<pre>srunch python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name clogged_lag_onestudy \ --study_count 1 \ --onehot-encoding \ --data-directory cloggedData \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \ --discard-features "time[s],AN_1_LL[m/s],AN_2_ LQ[m/s],AN_4_RR[m/s],ML_L L[mm],ML_LQ[mm],ML_RR[mm],L_wave_ht[mm],R_wave _ht[mm]" \ --feature-lag-amount 10 \</pre>	<pre>EX1: "mean_absolute_error": 8.405871253248808, "mean_absolute_percent_e rror": 22.38835712222203, "r_squared": 0.8954124825051717, "mean_squared_error": 155.1579462064207, "root_mean_squared_error ": 12.456241255146782 EX2: "mean_absolute_error": 2.1772749127126136, "mean_absolute_percent_e rror": 6.435187990637166, "r_squared": 0.9616884354713275, "mean_squared_error": 44.04433980268021, "root_mean_squared_error ": 6.6365909775034515</pre>

RQ only

Model	Arguments	Metrics
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RQ only clean data	<pre> srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name RQonly_noClog \ --study_count 100 \ --onehot-encoding \ --data-directory noClog \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s],AN_1_LL[m/s],AN_2_ LQ[m/s],AN_4_RR[m/s],ML_L L[mm],ML_LQ[mm],ML_RR[mm],L_wave_ht[mm],R_wave _ht[mm]" \ --sen-geometrical \ </pre>	EX1: <pre> "mean_absolute_error": 7.0968481786986315, "mean_absolute_percent_err or": 18.09182656214065, "r_squared": 0.8921572978743438, "mean_squared_error": 211.1932104882253, "root_mean_squared_error": 14.532488103839112, EX2: "mean_absolute_error": 6.553547641438189, "mean_absolute_percent_err or": 17.35619562201061, "r_squared": 0.9096767317259064, "mean_squared_error": 160.73556099857439, "root_mean_squared_error": 12.678152901687785, </pre>
RQ only clogged data	<pre> srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name RQonly_clog \ --study_count 100 \ --onehot-encoding \ --data-directory cloggedData \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s],AN_1_LL[m/s],AN_2_ LQ[m/s],AN_4_RR[m/s],ML_L L[mm],ML_LQ[mm],ML_RR[mm],L_wave_ht[mm],R_wave _h>" \ --sen-geometrical \ </pre>	EX1: <pre> "mean_absolute_error": 5.155936399569692, "mean_absolute_percent_err or": 11.427937723958143, "r_squared": 0.9112783915317112, "mean_squared_error": 131.62051154671073, "root_mean_squared_error": 11.47259829100238, EX2: "mean_absolute_error": 4.526015314401356, "mean_absolute_percent_err or": 11.533958466931004, "r_squared": 0.9042555371965918, </pre>

		<pre>"mean_squared_error": 110.07124626253196, "root_mean_squared_error": 10.491484464199141,</pre>
RQ only both	<pre>srun python run_study.py -- targets Count_EX1 Count_EX2 \ --study-name RQonly_both \ --study_count 100 \ --onehot-encoding \ --data-directory data \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s],AN_1_LL[m/s],AN_2_ LQ[m/s],AN_4_RR[m/s],ML_L L[mm],ML_LQ[mm],ML_RR[mm],L_wave_ht[mm],R_wave _h>" \ --sen-geometrical \</pre>	<pre>EX1: "mean_absolute_error": 6.213950907600255, "mean_absolute_percent_err or": 16.16286973878071, "r_squared": 0.8997334108763286, "mean_squared_error": 191.31749963134936, "root_mean_squared_error": 13.831756924966161, EX2: "mean_absolute_error": 5.549354963720419, "mean_absolute_percent_err or": 15.22087875623835, "r_squared": 0.9131364178553363, "mean_squared_error": 133.5086787870983, "root_mean_squared_error": 11.554595570036119,</pre>

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Last 200 rows remo ved	<pre>srun python run_study.py \ --targets Count_EX1 Count_EX2 \ --study-name 200rm \ --study_count 100 \ --onehot-encoding \ --data-directory bub_200_rm \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --discard-features "time[s]" \ --sen-geometrical \</pre>	

	<pre>--lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s],AN_3_RQ[m/s],AN_4_RR[m/s],ML_LL[mm],ML_LQ[mm],ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 10 \ --mould-position \</pre>	
Data cleani ng ignor e time sen geo mld- pos clog ged SEN with time lag 10 Binni ng	<pre>srunch python run_study.py --targets Count_EX1 Count_EX2 \ --study-name clogged_lag_binning_try2 \ --study_count 100 \ --onehot-encoding \ --data-directory cloggedData \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \ --lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s],AN_3_RQ[m/s],AN_4_RR[m/s],ML_LL[mm],ML_LQ[mm],ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 10 \ --binning-size 11</pre>	
Data cleani ng ignor e time sen geo mld- pos clog ged SEN with time lag 10 Try 2	<pre>srunch python run_study.py --targets Count_EX1 Count_EX2 \ --study-name clogged_lag_try2 \ --study_count 100 \ --onehot-encoding \ --data-directory cloggedData \ --subsample-shap \ --tree-method hist \ --clogging-factors \ --multithread \ --mould-position \ --discard-features "time[s]" \ --sen-geometrical \ --lagged-features "AN_1_LL[m/s],AN_2_LQ[m/s],AN_3_RQ[m/s],AN_4_RR[m/s],ML_LL[mm],ML_LQ[mm],ML_RQ[mm],ML_RR[mm]" \ --feature-lag-amount 10 \</pre>	